Ben M. Andrew

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Education

University of Manchester, PhD in Computer Science — Sep 2024 – Present

- Research focused on formal methods for understanding degraded systems.
- Supervised by Dr. Marie Farrell, Dr. Louise Dennis, and Prof. Michael Fisher.

University of Cambridge, BA (Hons) in Computer Science — Oct 2019 – Jun 2022

- Grade II.1 (70%).
- Dissertation titled *Delay-Tolerant Link-State Routing*. Implemented and empirically evaluated a failure-tolerant routing protocol in C to run on real Unix-based routers, comparing with traditional protocols. Supervised by Dr. Jackson Woodruff.

Publications

- Andrew, B. M., Dennis, L. A., Fisher, M., Farrell, M. *Counterexample-Guided Interval Weakening*. Rigorous State-Based Methods (ABZ) 2026. 10.1007/978-3-032-26752-8_1
- Andrew, B. M. *Weakening Goals in Logical Specifications*. Rigorous State-Based Methods (ABZ) 2025. 10.1007/978-3-031-94533-5_22

Industry Experience

Software Engineer, Tarides – Paris, France — Sep 2022 – Jun 2024

- Developed open-source continuous integration (CI) software for the OCaml language's ecosystem, involving systems programming to distribute dependency solving, compilation, and testing to many operating systems and architectures.
- Responsible for the CI of the Opam Repository, the central package universe for OCaml with tens of thousands of packages.

Embedded Systems Intern, Tarides – Paris, France — Jul – Sep 2021

- Research & development project to cross-compile the OCaml bytecode runtime to an ARM-based microcontroller using a real-time operating system as a base layer.

Computer Graphics Research Intern, University of Cambridge — Jun – Sep 2020

- Developed and implemented a visual model of ageing vision in virtual reality with Unity, as part of the Graphics and Displays Research Group.

Professional Activities

- Part of the local organising committee of Integrated Formal Methods (iFM) 2024.
- Member of the Autonomy and Verification Network.

Teaching Experience

University of Cambridge, Supervisor (TA equiv.) — Oct 2022 – Present

- Leading small group supervisions for undergraduate courses in computer science, including topics such as logic, formal verification, and semantics of programming languages.

University of Manchester, Graduate Teaching Assistant — Oct 2024 – May 2026

- Assisting in teaching graduate and undergraduate courses in computer science, including topics such as software engineering, logic, and formal verification.